

Hematopoiesis & Lymphoid System - Factoid Sheet

Week 7 Hematology • Lecture 3 • Adroja • memorize-first companion

ScholarRx: Erythropoiesis • Myeloid & Lymphoid Lineages • Anatomy of the Lymphatic System

Lecture map: development -> marrow factory -> B/T maturation -> node traffic -> MALT -> spleen blood filter -> thymus T-cell school.

Embryonic + adult hematopoiesis

When	Site	Memorize
Day 18 -> wk 8	Yolk sac	Primitive nucleated RBCs, embryonic Hb, macrophages
Wk 4-6	AGM	First true HSCs bud from hemogenic endothelium
Wk 6 -> birth	Liver/spleen	Liver dominates mid-gestation
Wk 10 -> life	Bone marrow	Dominant by ~wk 28; physiologic site after birth

Adult red marrow: vertebrae, sternum, ribs, pelvis, skull, proximal humerus/femur. Yellow marrow can reconvert under stress.

HSC + marrow

• HSC: CD34+ CD38- Lin- • Self-renewal • Multipotency • Quiescence • Rare, not identified by morphology alone	• Endosteum: quiescent niche • Cords: developing cells • Sinusoids: mature-cell exit • M:E ratio ~2:1-4:1
--	--

Lineage cold sets

Lineage	Sequence / fact	Driver
Erythroid	Proerythroblast -> basophilic -> poly -> ortho -> retic -> RBC	EPO
Neutrophil	Blast -> promyelo -> myelo -> metamyelo -> band -> seg	G-CSF
Platelet	Megakaryocyte endomitosis; 16N-64N; sheds into sinusoid	TPO

B cell: born + matures in marrow. **T cell:** born in marrow, matures in thymus.

Primary vs secondary lymphoid organs

Primary/generative	Secondary/peripheral
Bone marrow + thymus Production/education, antigen-independent	Nodes, spleen, tonsils, Peyer's, MALT Antigen encounter, clonal expansion, effector/memory

Histology + Traffic Factoids

Lymph node

Outside in: capsule -> subcapsular sinus -> cortex (B) -> paracortex (T + HEV) -> medulla -> hilum.

Zone	What lives there
Cortex	B-cell follicles / germinal centers
Paracortex	T cells + HEVs
Medullary cords	Plasma cells
Medullary sinuses	Macrophages + lymph flow

Two routes in: afferent lymphatics bring antigen/DCs; blood brings naive B/T through HEVs. One route out: efferent lymphatic at hilum.

MALT / tonsil / Peyer's

Peyer's patch

- Ileum
- Aggregated follicles
- Unencapsulated
- M cells sample lumen
- Secretory IgA output

Palatine tonsil

- Nonkeratinized squamous epithelium
- Deep crypts
- Follicles beneath surface
- Incomplete deep capsule

Spleen

Zone	Memorize
Red pulp	Cords of Billroth + sinusoids; tests/removes old RBCs
PALS	T-cell zone around central arteriole
Follicle	B-cell zone
Marginal zone	Macrophages + memory B; first blood-borne antigen contact

Spleen filters BLOOD. Open circulation: artery -> cords -> squeeze through sinus slits -> venous return. No spleen -> Howell-Jolly bodies + encapsulated-organism risk.

Thymus

- Anterior mediastinum
- Epithelium: **3rd pharyngeal pouch**
- Dark cortex, pale medulla
- Hassall corpuscles = medulla
- No follicles/germinal centers

- Positive selection = cortex:** can see self-MHC?
- Negative selection = medulla:** binds self too strongly?
- AIRE helps display tissue antigens
- Exit as CD4 or CD8 single-positive cells

If you see X -> think Y

Paracortex absent	T-cell deficiency / DiGeorge	HEV present	Lymph node / MALT, not spleen
Central arteriole in lymphoid tissue	Spleen	Hassall corpuscles	Thymus
Deep squamous crypts	Palatine tonsil	Ileal follicles + M cells	Peyer's patch
Howell-Jolly bodies	Asplenia/hyposplenia	Retic count rises	Marrow is responding

Last 5 minutes: Yolk -> AGM -> liver/spleen -> marrow. HSC CD34+ CD38- Lin-. Node cortex B/paracortex T. Spleen PALS T + follicle B + red-pulp RBC filter. Thymus positive cortex / negative medulla.

Primary source: Adroja 62-slide lecture. Official Week 7 LOs set emphasis. Companion is supplemental only; current-deck developmental timing is used when sources differ.